Big Ideas in Mathematics

Unpacking Big Idea about Proportionality

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_____ Orientations to Math Learning

There are two orientations to mathematics learning that are relevant to the design of the syllabuses. They are:

1. Learning mathematics as a tool - This orientation places emphasis on using mathematics as a tool to solve problems. The selection of content is based on its value in contexts where students are likely to apply mathematics.

2. Learning mathematics as a discipline - This orientation places emphasis on understanding the nature of mathematics. The selection of content is influenced by its value to illustrate the thinking and practices of mathematicians.

Every syllabus has elements of both orientations, but the balance between the two differs. For example, the design of the N(T)-Level Mathematics syllabus is influenced by a stronger orientation towards learning mathematics as a tool, whereas the design of the Additional Mathematics syllabuses is influenced by a stronger orientation towards learning mathematics as a discipline.

Nature of Mathematics

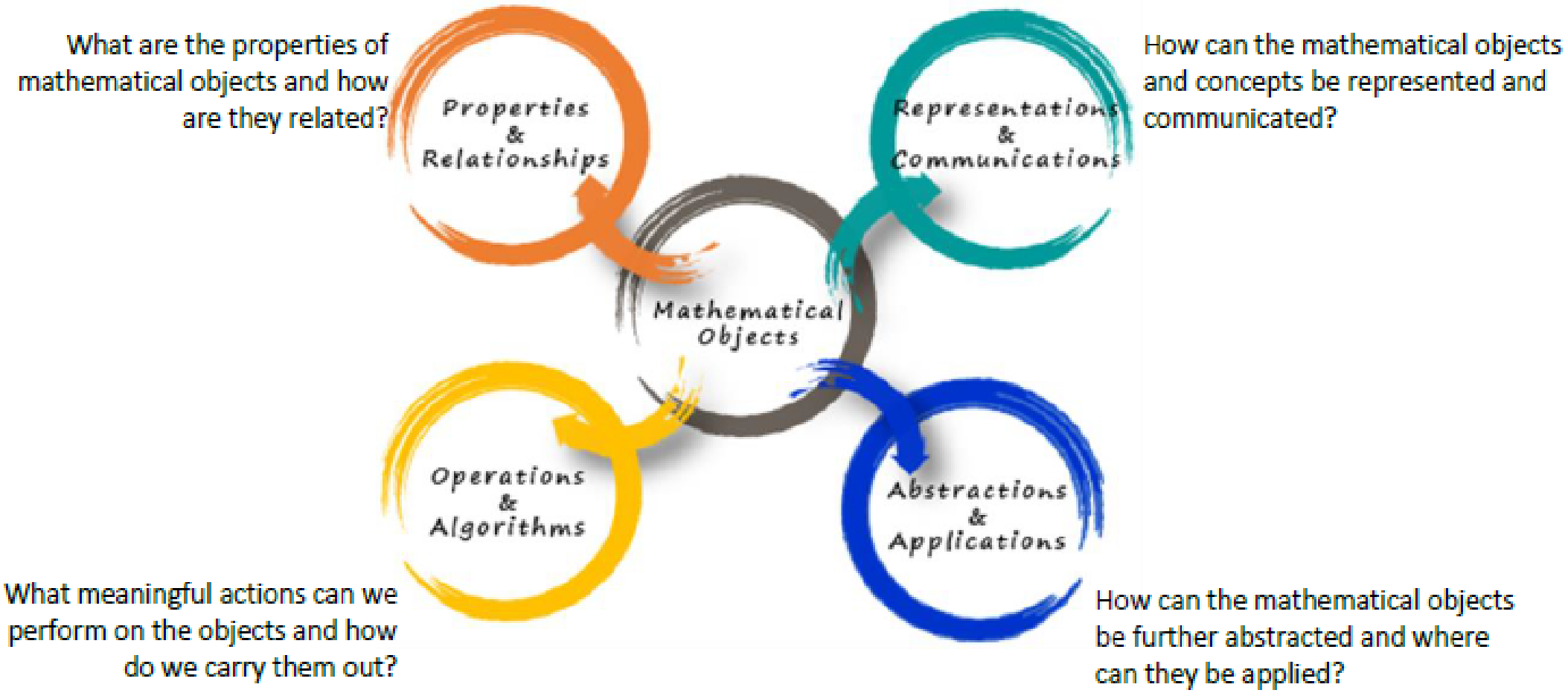
Nature of Mathematics

Mathematics can be described as a study of the *properties*, *relationships*, *operations*, *algorithms*, and *applications* of numbers and spaces at the very basic levels, and of abstract objects and concepts at the more advanced levels. Mathematical objects and concepts, and related knowledge and methods, are products of insight, logical reasoning and creative thinking, and are often inspired by problems that seek solutions. *Abstractions* are what make mathematics a powerful tool for solving problems. Mathematics provides within itself a language for *representing* and *communicating* the ideas and results of the discipline.

_____ in the study of mathematics:

- **P**roperties and Relationships (concepts)
- **O**perations and Algorithms (skills)
- **R**epresentations and Communications (processes)
- **A**bstractions and Applications (processes)

Four Recurring Themes in Study of Maths



Big Ideas in Mathematics

Big ideas express ideas that are central to mathematics. They appear in different topics and strands. There is a continuation of the ideas across levels. They bring coherence and show connections across different topics, strands and levels. The big ideas in mathematics could be about one or more themes, that is, it could be about *properties and relationships* of mathematical objects and concepts and the *operations and algorithms* involving these objects and concepts, or it could be about *abstraction and applications* alone. Understanding the big ideas brings one closer to appreciating the nature of mathematics.

Eight clusters of big ideas are listed in this syllabus. These are not meant to be authoritative or comprehensive. They relate to the four themes that cut across and connect concepts from the different content strands, and some big ideas extend across and connect more concepts than others. Each cluster of big ideas is represented by a label e.g. big ideas about Equivalence, big ideas about Proportionality, etc.

Eight Clusters of Big Ideas (for Sec)

- **If DN, PM me**
- If _____, private message me
- Invariance, Functions, Diagrams, Notations, Proportionality, Measures, Models, Equivalence
- For **primary school**, only **6 clusters of big ideas**: exclude functions and models (model method is under diagrams); **IMPEND** (about to happen)

Teaching towards Big Ideas

- "A big idea is a _____ **of an idea** that is central to the learning of mathematics, one that links numerous mathematical understandings into a coherent whole (Charles, 2005, p. 10)
- Proportionality is not the big idea: not big idea of proportionality, but big idea **about** proportionality
- What big idea can you say _____ proportionality?
  - "Proportionality is a relationship between two quantities that allows one quantity to be computed from the other based on multiplicative reasoning" is a **statement** of a big idea **about** proportionality

Charles, R. I. (2005). Big ideas and understandings as the foundation for elementary and middle school mathematics. *Journal of Mathematics Education Leadership, 2005, 8*(1), 9-24.

Statements about Proportionality

Big Ideas about Proportionality

Main Theme: Properties and Relationships

Proportionality is a relationship between two quantities that allows one quantity to be computed from the other based on multiplicative reasoning. Fraction, ratio, rate and percentage are different but related mathematical concepts for describing the proportional relationships between two quantities that allow one quantity to be computed from the other related quantity. In geometry, proportional relationships undergird important concepts such as similarity and scales. In statistics, proportional relationships are the basis for constructing and interpreting many statistical diagrams such as pie charts and histograms. Underlying the concept of proportionality are two quantities that vary in such a way that the ratio between them remains a constant.

Teaching towards Big Ideas

- It is not teaching about big ideas, but **teaching** _____ big ideas in mathematics by **forming useful statements** about a big idea
- **Not helpful** to just say, e.g. this is a case of proportionality: we need to say more, e.g. what is proportional to what? (more later)
- Proportionality is **not a big idea** but a _____ of big ideas because there are many big ideas about proportionality
- MOE syllabus document listed 8 **clusters** of big ideas

Proportionality

- Unpacking big idea about proportionality: connecting ratio, rate and proportion (and implications for teaching)
- For a teacher, **content knowledge** is as important as **pedagogical content knowledge** (Ball, Thames & Phelps, 2008; Shulman 1986)
  - So that we do not teach wrong things (difficult for students to un-learn)
- As a teacher, we should also _____ than what our students need to learn so that we can address their questions that may be beyond the syllabus
  - But we have to **decide when and what** to tell our students

Ball, D. L., Thames, M. H., & Phelps, G. (2008). Content knowledge for teaching: What makes it special? *Journal of Teacher Education, 59*, 389-407.
Shulman, L. S. (1986). Those who understand: Knowledge growth in teaching. *Educational Researcher, 15*(2), 4-14.

Why proportionality?

- "**Proportional reasoning** is at the core of so many important concepts in **mathematics** and **science**, including similarity, relative growth and size, dilations, scaling, pi, constant rate of change, slope, _____, percent, trig _____, probability, relative frequency, density, and direct and inverse variations." (Heinz & Sterba-Boatwright, 2008, p. 528)
- One of the most commonly applied mathematics in the **real world** (Lanius & Williams, 2003)
- But research has found that many students and even **adults** have difficulties in reasoning proportionally (Behr, Harel, Post & Lesh, 1992)

Behr, M., Harel, G., Post, T., & Lesh, R. (1992). Rational number, ratio, and proportion. In D. Grouws (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 296–333). New York: McMillan.

Heinz, K., & Sterba-Boatwright, B. (2008). The when and why of using proportions. *Mathematics Teacher, 101*, 528-533.

Lanius, C. S., & Williams, S. E. (2003). Proportionality: A unifying theme for the middle grades. *Mathematics Teaching in the Middle School, 8*, 392-396.

Why proportionality?

- Many **students** are unable to distinguish "between and among concepts of decimal, percent, ratio, proportion, and proportional reasoning" (Ojose, 2015, p. 111)
- May be because of the way topics are sequenced by teachers during instruction without presenting the opportunity for students to see the _____ and the **interconnectedness** among the various mathematical concepts (ibid.)
- So the question is how are ratio, rate and proportion related?

Ojose, B. (2015). Proportional reasoning and related concepts: Analysis of gaps and understandings of middle grade students. *Universal Journal of Educational Research, 3*, 104-112.

Comparative Situation

- 30 boys and 10 girls in a class: How many more boys are there than girls?
- There are 20 **more** boys than girls (**absolute / additive terms**)
- There are 3 **times** as many boys as girls (**relative / terms**)
- No. of boys : no. of girls = 3 : 1 (relative / multiplicative)
- Students tend to see comparative situations in absolute (additive) rather than relative (multiplicative) terms (Fielding-Wells, Dole, & Makar, 2014), partly because it is easier to see '20 more boys' than '3 times as many boys'

Fielding-Wells, J., Dole, S., & Makar, K. (2014). Inquiry pedagogy to promote emerging proportional reasoning in primary students. *Mathematics Education Research Journal, 26*, 44-47.

Ratio

- **Ratio compares** the **relative sizes** of two or more quantities of the **same kind**
- No. of boys : no. of girls = 3 : 1 is a ratio because we are comparing _____ **(same kind)**, not comparing boys and girls (children)
- In algebra: x boys and y girls: x and y are numbers
- **Not "ratio of boys to girls"** but "ratio of no. of boys to no. of girls"
- If quantities of same kind are just **numbers**, they have no units

Ratio: Same kind

- What if quantities of same kind are **measurements**, must they have same units? Can we write a map scale as 1 cm : 100 m?
- Ratio notation $a : b$ has no units, so map scale is 1 : 10 000
- **Do not write 1 cm : 100 m**, but write _____
- Does this mean if two measurements have different units (e.g. 1 cm and 100 m), it is not a ratio?
- May be helpful to distinguish between **concept of ratio** and **ratio** _____ $\boldsymbol{a : b}$
- If two measurements have same type of units (that can be converted from one to the other), we can still use **ratio** to compare, but if we write it using **ratio notation**, it must not have units **(convention)**

Ratio Notation

- **Usual textbook definition:** $a : b$, where $b \neq 0$
- Why $b \neq 0$ is because $a : b$ can be written as $\frac{a}{b}$
- Can $a = 0$?
  - E.g. no. of boys : no. of girls = 0 : 40
  - No. of girls : no. of boys = 40 : 0?
  - Is 0 : 40 the same as 0 : 20 or 0 : 1?
- If a quantity is 0, _____ to use ratio as a basis of comparison with another quantity: we just say that the quantity is 0

Ratio Notation

- **Usual textbook definition:** $a : b$, where $b \neq 0$
- Can a or b be negative?
  - Most (if not all) **scalar** quantities in real world are non-negative
  - Ideal gas law: $PV = nRT$, where T is absolute temperature (measured in Kelvin, never in °C or °F which are arbitrary) and $T > 0$ (physically impossible to reach 0 Kelvin)
  - Any textbook questions: $a, b > 0$
- **New definition of ratio notation:** $a : b$, where _____ **(now in both 2020 Sec 1 textbooks)**

Key Takeaway #1: Ratio (Summary)

- **Ratio** is used to compare the _____ of **two or more** quantities of the **same kind**
- _____ means the quantities are either:

  (i) **numbers** (of objects or people or animals), or

  (ii) **measurements** with same type of units (that can be converted from one to the other, e.g. cm and metres; not cm and grams)

- Ratio **notation** $a : b$ (or $a : b : c$ etc.) has **no units**

Comparing different kinds

- If two quantities are of _____, can we use ratio to compare? E.g. 5 pens : $10?
- Does it make sense to compare **relative sizes (ratio)** of the no. of pens and the cost of pens?
- We can compare relative sizes of no. of blue pens and no. of red pens; or relative sizes of cost of a blue pen and cost of a red pen
- We will use another way to compare no. of pens and the cost of pens, which is **rate**

Rate: Per unit of another quantity?

- Usual definition: Rate can be expressed as one quantity **per unit of another quantity**. But are there exceptions?
- E.g. an item costs \$2 per _____; or a smartphone plan costs \$10 per _____ of data
- Rate can be expressed as one quantity **per *n* units of another quantity**, where *n* is some 'nice' number
- Second quantity **does not have to be time**
- Above examples apply if **rate $\frac{y}{x}$ is constant (one idea of rate)**

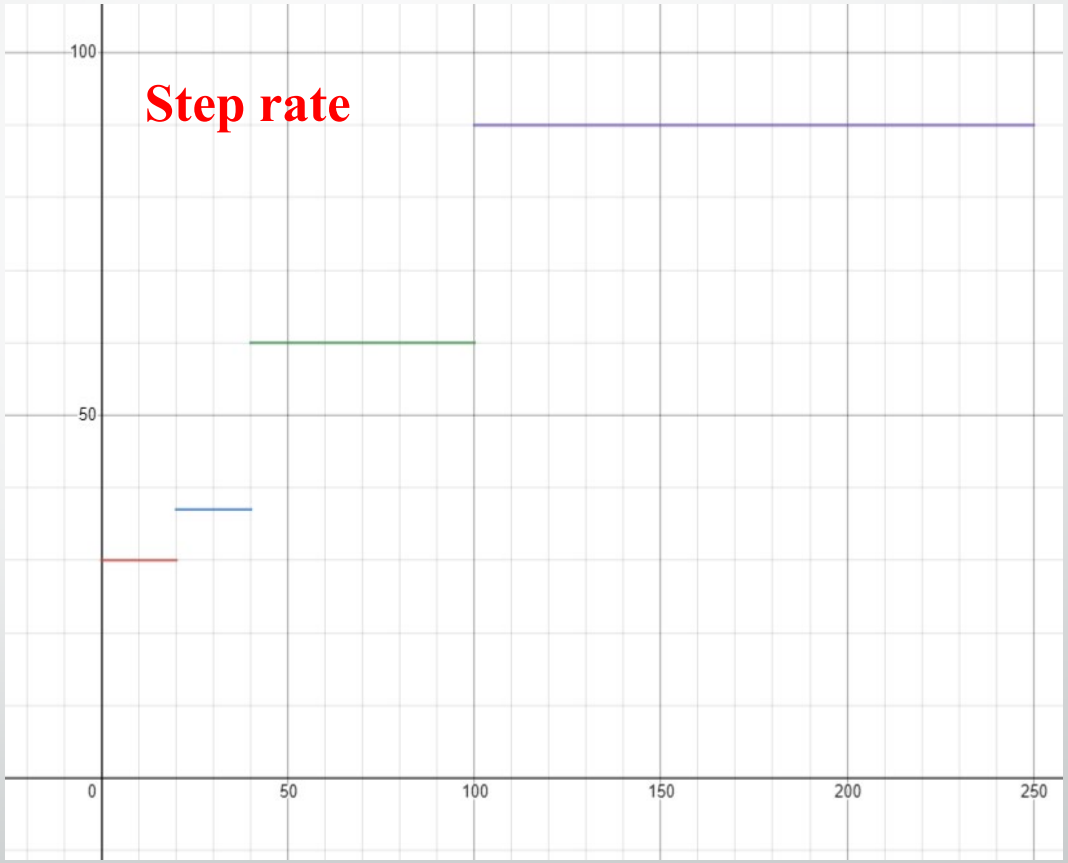
Is this idea of rate comprehensive enough?

What kind of rate is this?

- 2016 O-level Exam Paper 2 on Problem in Real-World Contexts

LOCAL MAIL RATES (i

For Letters, Postcards, Printed Papers and Packets/Packages		
Weight Step Up To	Standard Mail	Non-Standard Mail
Regular		
20 g	$0.30	$0.60
40 g	$0.37	
Large		
100 g	$0.60	$0.90
250 g	$0.90	$1.15
500 g	$1.15	$1.70
1 kg	$2.55	
2 kg	$3.35	



Progressive income tax rates in Singapore (**not step rate**, but it has different income brackets)

Rate of Change (Add. Maths)

- Add. Maths Calculus: **Rate of change** of y w.r.t. x (i.e. $\frac{\mathbf{d}y}{\mathbf{d}x}$)
- For straight line $y = kx + c$, gradient $k = \frac{\mathrm{d}y}{\mathrm{d}x}$ = **constant**
- But **rate** $\frac{y}{x} = k + \frac{c}{x}$ is _____, unless $c = 0$
- If $c = 0$, rate $\frac{y}{x} = k$ is constant, i.e. $y = kx$
- For $y = kx$ (straight line that passes through origin), $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x} = k$ is a constant
- **Rate** $\frac{y}{x}$ compares how y changes with x; **rate of change** $\frac{\mathrm{d}y}{\mathrm{d}x}$ compares how Δy (change in y) changes with Δx (change in x)

Constant Rate vs. Constant Rate of Change

- In general, **gradient of line = $\frac{\mathrm{d}y}{\mathrm{d}x}$ (rate of change)**, not $\frac{y}{x}$ (rate), unless the line passes through the origin
- In Add. Maths, when rate of change $\frac{\mathrm{d}y}{\mathrm{d}x}$ is constant, we also say that rate is constant!
- But rate in O-level Maths is $\frac{y}{x}$, which is constant for most cases
- Perhaps when $\frac{\mathrm{d}y}{\mathrm{d}x}$ is constant, we should be more specific to say that rate _____ is constant because 'constant rate' can also mean rate $\frac{y}{x}$ is constant; and $\frac{\mathrm{d}y}{\mathrm{d}x} \neq \frac{y}{x}$ in general

Is an exchange rate a constant rate?

- What is not constant is how an exchange rate changes with time: **rate of change of exchange rate with _____** is not constant
- But an exchange rate is a _____ **rate** at a given period or time (e.g. at money changer)
  - US$1 = S$1.37
  - US$2 = S$2.74 (rate is still the same: US$1 = S$1.37)
  - US$0.50 = S$0.685 (rate is still the same: US$1 = S$1.37)
- "Is an exchange rate constant?" is vague because it can mean:
  - "Is an exchange rate constant with respect to time?"
  - "Is an exchange rate a constant rate?"

Rate: Different Kinds?

- Most textbooks use rate to compare two quantities of **different kinds**
- Can rates be used to compare two quantities of _____ _____?
- Unemployment rate = no. of unemployed people divided by no. of working people, and then expressed as a percentage
- E.g. Singapore's unemployment rate in 2024 = 2.8%, i.e. 2.8 unemployed people per 100 working people, or 28 unemployed people per 1000 working people

Rate: Comparing how many quantities?

- Can we use a rate to compare three or more quantities?
- Additional Mathematics: $\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t}$
- But each of three rates of change still between _____ **quantities**
- We **cannot** use ***one*** rate (or rate of change) to compare three quantities because we cannot have dy/dx/dt, but we need to have **three connected rates of change** to compare the three quantities: $\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t}$

Key Takeaway #2: Rate (Summary)

- **Rate** is a **measure** of how one quantity _____ with another (it does not matter whether the **two** quantities are of **same kind or of different kinds**)
- **Rate $\frac{y}{x}$** (how **y changes with x**): If rate $\frac{y}{x}$ is constant, it can be expressed as a quantity per n units of another quantity
- If rate $\frac{y}{x}$ is not constant, we can look at other types of rates:
  - **Rate of change $\frac{dy}{dx}$**: can be constant or variable
  - Other variable rates such as step rates, progressive income tax rates, irregular rates (e.g. heartbeat)

Ratio vs. Rate

- Some overseas textbooks (e.g. Cathcart, Pothier, Vance & Bezuk, 2016; Van de Walle, Karp & Bay-Williams, 2010): a rate is a ratio
- A square is a rectangle means a square **must** _____ all the properties of a rectangle; so if a rate is a ratio, then you must be able to use the ratio notation $a : b$ (no units) for rates as well
- But those textbooks did not do so; then how can a rate be a ratio?
- Local textbooks: clear distinction between ratio and rate, but are they related?

Cathcart, W. G., Pothier, Y. M., Vance, J. H., & Bezuk, N. S. (2006). *Learning mathematics in elementary and middle schools* (4th ed.). Upper Saddle River, NJ: Pearson.

Van de Walle, J. A., Karp, K. S., & Bay-Williams, J. M. (2010). *Elementary & middle school mathematics* (7th ed.). Boston, MA: Pearson.

Ratio vs. Rate

- Math Snacks: Bad Date (Video): https://www.youtube.com/watch?v=BZ1M01YBKhk
- Ratio of woman's no. of words to man's no. of words was 25 : 175 or 1 : 7
- Then the woman said, "For every word I spoke, he spoke 7 words."
- Rate: "He spoke 7 words **per** word that I spoke."
- **Ratio (of two quantities) can be _____ to a rate**
- Can you always do that?
- E.g. if ratio of no. of boys to no. of girls is 2 : 1, can you say that there are 2 boys per girl? Does that make sense to you?

List of countries by sex ratio

From Wikipedia, the free encyclopedia

The human sex ratio is the number of males for each female in a population. This is a list of sex ratios by country or region.

Country/region	at birth (CIA estimate, 2016)[1]	0–14 years	15–24 years	25–54 years	55–64 years	over 65	total	at birth (WDB estimate, 2012)[5]
– *World*	1.09	1.07	1.07	1.02	0.95	0.805	1.015	1.07
Afghanistan	1.05	1.03	1.04	1.04	0.97	0.86	1.03	1.07
Albania	1.10	1.12	1.07	0.91	0.98	0.89	0.98	1.07
Singapore	1.07	1.05	0.97	0.95	1.00	0.83	0.96	1.07

- Singapore: 1.07 males per female (idea of rate); ratio of no. of males to no. of females = 1.07 : 1 (or 107 : 100)
- Writing it as **rate** makes it easier to **compare** across different groups (or countries) because bases are the same (= 1)

Ratio vs. Rate

- Which class has a **higher proportion** of boys? (***not*** which class has more boys because total no. of students in each class may be different)

Class	A	B	C
No. of Boys to No. of Girls (Ratio)	3 : 2	5 : 2	3 : 1

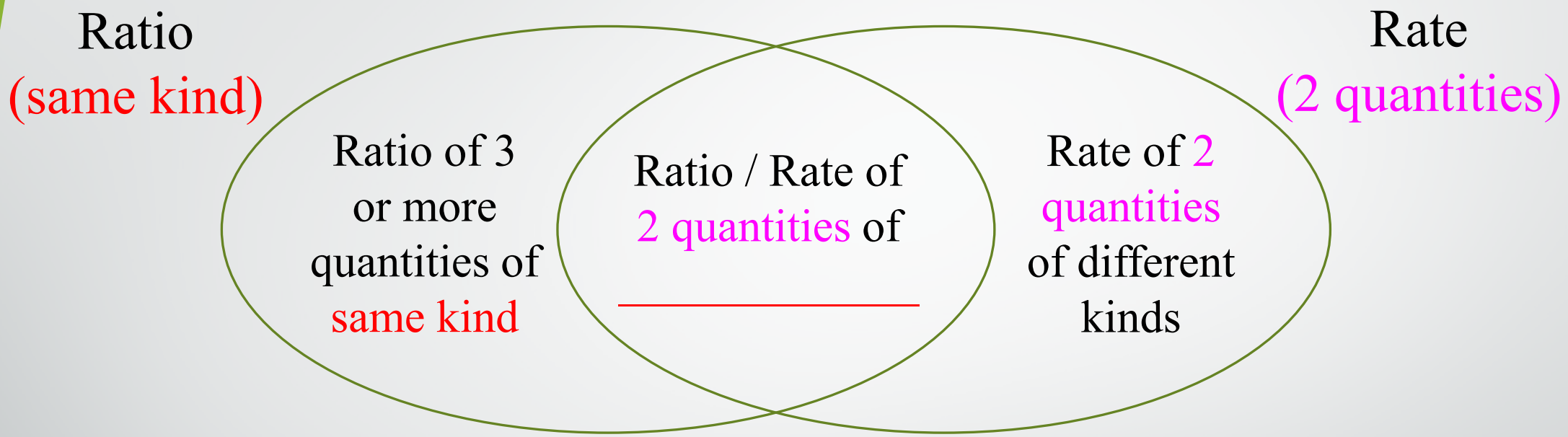
- Can easily tell Class B has a higher proportion of boys than Class A (because both ratios have _____)
- For Class A and Class C, have to reason a bit
- For Class B and Class C, even harder to tell

Class	A	B	C
No. of boys per girl (Rate)	1.5	2.5	3

Ratio vs. Rate

- **Ratio (of _____ quantities) can be converted to a rate**, simply because ratio $a : b$ **can be written as $\frac{a}{b}$ (fraction)**, and $\frac{a}{b}$ is idea of rate, e.g. ratio 107 : 100 can be written as $\frac{107}{100}$ (107 males per 100 females) or $\frac{1.07}{1} = 1.07$ (1.07 males per female)
- Ratio of **three or more quantities cannot** be converted to a rate, because, e.g. $a : b : c$ cannot be expressed as a fraction, which is why rate is used to compare two quantities only
- Rate comparing two quantities of **different kinds cannot** be converted to a ratio, because latter is used to compare relative sizes of quantities of same kind

Key Takeaway #3: Ratio vs. Rate (Summary)



- For **two quantities of same kind**, ratio can be converted to rate and vice versa
- Can either use ratio or rate (or both) to compare, but sometimes, it may be more **meaningful or** _____ to use one over the other

Some Implications for Teaching

- **Give** _____ **examples** when introducing a concept
  - E.g. do we want to give an example of a rate that compares two quantities of same kind when introducing what is a rate?
  - Some students may say this is ratio: do we want confusion at this stage?
- **Give** _____ **examples** to cover all cases
  - Do we want to give examples of rates involving only two quantities of different kinds?
  - Start with different kinds, but must also cover same kind: this can lead to discussion of which rates can be converted to ratios

Some Implications for Teaching

- Or do we want to teach our students ratio and rate in such a way that these two concepts are not related to each other at all? We want students to see _____ across topics/concepts
- More importantly, do we want to teach our students when it is **more meaningful or helpful** to use one (i.e. ratio or rate) over the other when making comparison?
- Don't confuse Lower Sec students about rate of change, but for Add. Maths students, do we want to help them see that, in general, **rate of change $\frac{dy}{dx}$ is _____ equal to rate $\frac{y}{x}$** which they have learnt in Lower Sec?

Proportion: Different Meanings

proportion *noun* **(IMPORTANCE)**

[U] **used in a number of phrases to mean importance and seriousness:**

*You have to keep a **sense of** proportion (= the ability to understand what is important and what is not).*

*I think a certain amount of worry about work is very natural, but you've got got to **keep it in** proportion (= judge correctly its seriousness).*

proportion *noun* **(SIZE)**

C2 [C or U] **the correct or most attractive relationship between the size of different parts of the same thing or between one thing and another:**

*Your legs are very much **in** proportion **to** (= the right size for) the rest of your body.*

*His feet seem very small **in** proportion **to** his body.*

*My head was much nearer the camera than the rest of me so I'm all **out of** proportion.*

Proportion: Different Meanings

proportion *noun* **(AMOUNT)**

C1 [C, + sing/pl verb] **the number or amount of a group or part of something when compared to the whole:**

*Children make up a large proportion **of** the world's population.*

*A higher proportion **of** men are willing to share household responsibilities than used to be the case.*

*The report shows that poor families spend a larger proportion **of** their income on food.*

- **Proportion** can refer to a **part-whole ratio**

Proportion as Part-Whole Ratio

- Which class has a **higher proportion** of boys?

Class	A	B	C
No. of Boys to No. of Girls (Ratio)	3 : 2	5 : 2	3 : 1
No. of boys per girl (Rate)	1.5	2.5	3

- But **proportion** of boys in, e.g. Class A = 3 : 5 or $\frac{3}{5}$ or 60% **(part-whole ratio)**
- Can we ask what is the proportion of boys to girls in Class A?
- This is just ratio of no. of boys to no. of girls **(part-part ratio)**

2020 MOE Syllabus Document

8.	The diagram shows two boxes of cereals. Cereal A Contains 8.5% fruit Cereal B 7 g of fruit in every 75 g portion Which box of cereal has a greater proportion of fruit and by how much?	AO 3 This question requires students to *provide mathematical explanation in the context of a given problem.*

Proportion vs. Proportionality

- This more mathematical idea of **proportion** (part-whole ratio or part-part ratio) is still different from direct or inverse proportion (which is _____ between two quantities)
- E.g. proportion of boys in the class is 3 : 5, or proportion of boys to girls is 3 : 2
- If we let y be the no. of boys and x be the no. of girls, then $y : x = 3 : 2$ is a statement of **proportionality** --- relationship between the two quantities x and y
- In this case, the relationship is direct proportion
- Usually, 'is proportional to' refers to direct proportion unless we say 'is inversely proportional to' (if in doubt, state 'is directly proportional to')

Direct Proportion

- Library fines for one overdue book:

× 3 × 2 × 2 ÷ 2 or × 0.5

No. of days (x)	1	2	3	4	5	6	7	8
Fine (y cents)	15	30	45	60	75	90	105	120

× 2 × 3 × 2 ÷ 2 or × 0.5

- When x is doubled, y is _____; when x is halved, y is halved; etc.
- **Concept of direct proportion:** when x is multiplied by n, y is also multiplied by n (n can be 0.5 etc.); **multiplicative (not additive) reasoning**

Not Additive Reasoning

- Some students may use additive reasoning like this:

+ 1 + 1

No. of days (x)	1	2	3	4	5	6	7	8
Fine (y cents)	15	30	45	60	75	90	105	120

+ 15 + 15

- Although it works here, it is **not enough** for this to be direct proportion

+ 1 + 1

No. of days (x)	1	2	3	4	5	6	7	8
Fine (y cents)	20	35	50	65				

+ 15 + 15

But in this table, when x is doubled, y is **not** doubled (_____ **direct proportion)**

Multiplicative Reasoning

Big Ideas about Proportionality

Main Theme: Properties and Relationships

Proportionality is a relationship between two quantities that allows one quantity to be computed from the other based on multiplicative reasoning. Fraction, ratio, rate and percentage are different but related mathematical concepts for describing the proportional relationships between two quantities that allow one quantity to be computed from the other related quantity. In geometry, proportional relationships undergird important concepts such as similarity and scales. In statistics, proportional relationships are the basis for constructing and interpreting many statistical diagrams such as pie charts and histograms. Underlying the concept of proportionality are two quantities that vary in such a way that the ratio between them remains a constant.

Direct Proportion

x_1 $\times 2$ x_2 $\times 3$

No. of days (x)	1	2	3	4	5	6	7	8
Fine (y cents)	15	30	45	60	75	90	105	120

y_1 $\times 2$ y_2 $\times 3$

x_2 and y_2 can represent any pair of values: no need to use x_3 and y_3

- $\frac{x_2}{x_1} = 2$ and $\frac{y_2}{y_1} = 2$, so $\frac{y_2}{y_1} = \frac{x_2}{x_1} = 2$
- $\frac{x_2}{x_1} = 3$ and $\frac{y_2}{y_1} = 3$, so $\frac{y_2}{y_1} = \frac{x_2}{x_1} = 3$
- "For a mathematician, a **proportion** is a statement of **equality of two ratios**, i.e. $a/b = c/d$." (Tourniaire & Pulos, 1985, p. 181)
- But the **two ratios are** _____ **constant** (even for same scenario of library fines)

Tourniaire, F., & Pulos, S. (1985). Proportional reasoning: A review of the literature. *Educational Studies in Mathematics, 16*, 181-204.

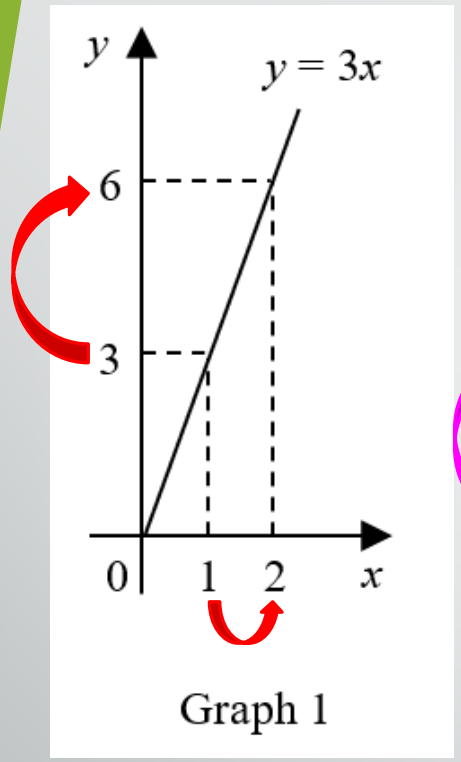
Key Takeaway #4: Direct Proportion

- Two related ideas of direct proportion:
  - **(Direct) Proportional Reasoning:** When one quantity (or variable) is multiplied by *n*, the other quantity is also multiplied by *n*
  - **Equality of two ratios: ——— = ———**

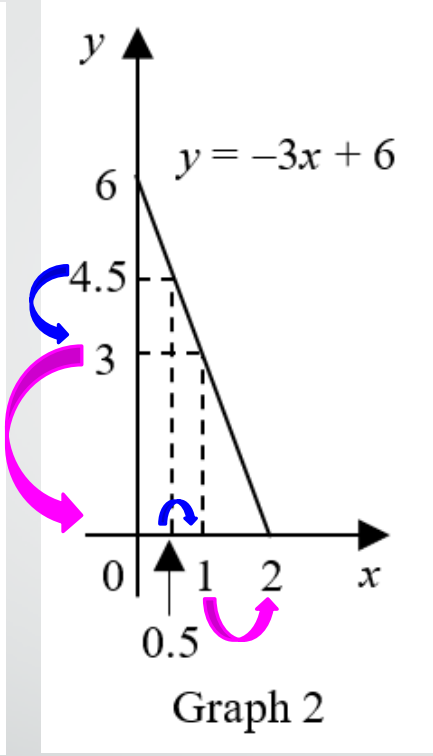
Direct Proportion

- Can we say, "For direct proportion, when x increases, y increases"?
  - **Other types of increase**, e.g. quadratic or exponential increase
  - Important to say, "When x increases, y increases _____, and **by 'proportionally', we mean** when x is doubled, y is also doubled; when x is halved, y is also halved; etc." (easier for students to understand)
  - Or "… and by 'proportionally', we mean when x is multiplied by n, y is also multiplied by n" (too formal for students)

Graphical Representations



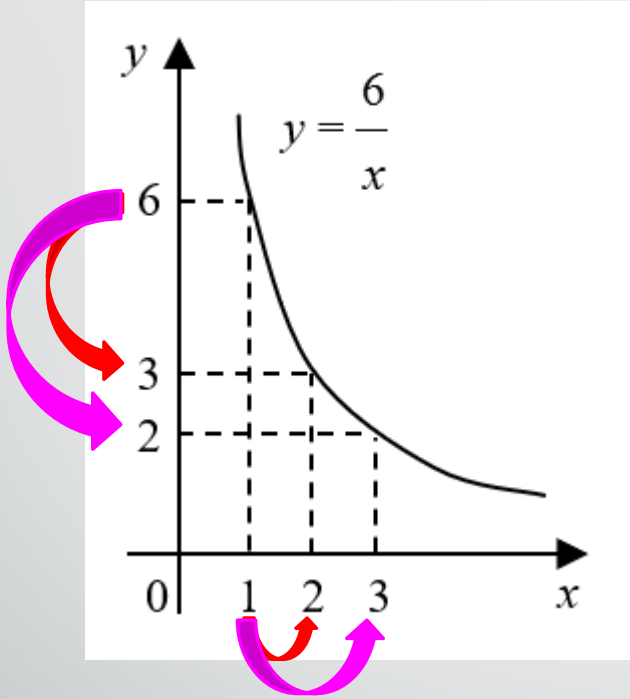
Graph 1



Graph 2

- Graph 1: Direct proportion between x and y (when x is doubled from 1 to 2, y is doubled from 3 to 6)
- Graph 2: Inverse proportion?
- When x is doubled from 1 to 2, y decreases from 3 to 0 (but not halved)
- When x is doubled from 0.5 to 1, y decreases from 4.5 to 3 (still not halved)
- **Graph 2 is not inverse proportion**: x and y are _____ **related**

Graphical Representations



- Is this inverse proportion?
- When x is **doubled** from 1 to 2, y is _____ from 6 to 3
- When x is multiplied by 3 from 1 to 3, y is divided by 3 from 6 to 2
- This is **inverse proportion**

Direct Proportion

No. of days (x)	1 (x_1)	2 (x_2)	3	4	5	6	7	8
Fine (y cents)	15 (y_1)	30 (y_2)	45	60	75	90	105	120
Rate $\frac{y}{x}$ (cents/day)	15	15	15	15	15	15	15	15

- $\frac{y}{x} = k$, where k is a non-zero constant
- $y = kx$, where $k \neq 0$
- $\frac{y_2}{x_2} = \frac{y_1}{x_1}$ **(equality of 2 rates)** can be obtained from $\frac{y_2}{y_1} = \frac{x_2}{x_1}$ **(equality of 2 ratios)** by some manipulations
- But the **rate** is a _____: $\frac{y_1}{x_1} = \frac{y_2}{x_2} = \frac{y_3}{x_3} = \ldots = k$ **(equality of rates)**
- Unlike **ratio**, which is **not a constant**, e.g. $\frac{y_2}{y_1} = \frac{x_2}{x_1} = 2$, or $\frac{y_2}{y_1} = \frac{x_2}{x_1} = 3$, etc.

x_2 and y_2 can represent any pair of values: no need to use x_3 and y_3

Direct Proportion

x_1 $\times 2$ x_2 $\times 3$

No. of days (x)	1	2	3	4	5	6	7	8
Fine (y cents)	y_1 15	30 y_2	45	60	75	90	105	120
Rate $\frac{y}{x}$ (cents/day)	15	15	15	15	15	15	15	15

$\times 2$ $\times 3$

- **Rate $\frac{y}{x} = \frac{y_1}{x_1} = \frac{y_2}{x_2} = \frac{y_3}{x_3} = \ldots$ = 15 (constant of proportionality)**
- **Ratios**: $\frac{y_2}{y_1} = \frac{x_2}{x_1}$ **= 2 (same multiplier)**, or $\frac{y_2}{y_1} = \frac{x_2}{x_1}$ **= 3** (same multiplier) (but multiplier is _____)
- **Importance** of distinguishing between ratio and rate: Which number is ratio and which number is rate? **Which one is constant?**

x_2 and y_2 can represent any pair of values: no need to use x_3 and y_3

Direct Proportion vs. Constant Rate

- Does constant rate imply direct proportion?
- Which constant rate are we referring to?
- If straight line $y = kx + c$, then constant rate refers to **constant rate of change** $\frac{\mathbf{d}y}{\mathbf{d}x}$ (or $\frac{\Delta y}{\Delta x}$), which is _____: need $c = 0$
- Sec 2: Straight line must pass through origin (or **both quantities are 0 at the same time**)
- Sec 1: Constant rate refers to **rate $\frac{y}{x}$ is constant**, then it's direct proportion: _____ to check both quantities are 0 at the same time (although latter can be helpful to check if $\frac{y}{x}$ is constant)

Direct Proportion vs. Linearity

- If linear relation $y = kx + c$, then y is directly proportional to x if and only if $c = 0$
- **Direct proportion** is a _____ **of linear relation**
- Inverse proportion $y = \frac{k}{x}$ is very different from linear relation

Key Takeaway #5 (Equivalent Statements)

- y is directly proportional to x
- Equality of two ratios: —— = —— for any two pairs (x_1, y_1) & (x_2, y_2)
- When x is multiplied by n, y is also _____
- Equality of two rates: —— = —— for any two pairs (x_1, y_1) & (x_2, y_2)
- Rate $\frac{y}{x}$ is _____

Teaching towards big idea: forming statements about (direct) proportionality

- Rate of change $\frac{\mathrm{d}y}{\mathrm{d}x}$ (or $\frac{\Delta y}{\Delta x}$) is constant, and both quantities x and y are _____ at the same time

Inverse Proportion

- $\boldsymbol{y = \frac{k}{x}}$ or _____, where k is a non-zero constant
- $\frac{y_2}{x_1} = \frac{y_1}{x_2}$ **(equality of 2 rates)** can be obtained from $\frac{y_2}{y_1} = \frac{x_1}{x_2}$ **(equality of 2 ratios)** by some manipulations
- But unlike direct proportion, for **inverse proportion**, the **rate is not a constant**; what is **constant** is the **product** of the two quantities (or variables)

Key Takeaway #6: Direct & Inverse Proportions

Direct Proportion

- When one quantity is **multiplied** by *n*, the other quantity is also **multiplied** by *n*
- **Equality of two ratios** $\frac{y_2}{y_1} = \frac{x_2}{x_1}$
- Equality of two (or more) rates $\frac{y_2}{x_2} = \frac{y_1}{x_1}$
- **Rate** $\frac{y}{x}$ **is constant**

Inverse Proportion

- When one quantity is **multiplied** by *n*, the other quantity is _____ by *n*
- **Equality of two ratios** $\frac{y_2}{y_1} = \square$
- Equality of two rates $\frac{y_2}{x_1} = \frac{y_1}{x_2}$
- _____ **is a constant**

Proportionality in the real world

- "Although most people are probably **unaware** of the mathematical definition of proportions [i.e. **equality of two ratios**], they do use them in familiar situations." (Tourniaire & Pulos, 1985, p. 181)
- How people use proportions is via the idea of **proportional** _____: when one quantity is multiplied by *n*, the other quantity is multiplied (or divided) by *n*, depending whether it is direct or inverse proportion
- **Don't need (equality of) ratios** to teach proportional reasoning

Tourniaire, F., & Pulos, S. (1985). Proportional reasoning: A review of the literature. *Educational Studies in Mathematics, 16*, 181-204.

Primary 5 Textbook under Rate

- Mickey saves \$72 in 18 days. He saves the same amount each day. How many days will Mickey take to save \$160? (**Primary 5** Textbook, 2017, p. 74)

$$\$72 \rightarrow 18 \text{ days}$$

$\div 72$ $\qquad$ $\div 72$

$$\$1 \rightarrow \frac{18}{72} \text{ day}$$

$\times 160$ $\qquad$ $\times 160$

$$\$160 \rightarrow \frac{18}{72} \times 160 \text{ days}$$

- _____ **method** makes use of **proportional reasoning**
- Not natural to use equality of two ratios or form equation $y = kx$ to solve

2020 Sec 1 Textbook p. 49 (under Rate)

Worked Example 9

PPT *

PPT *

Rates involving quantities of different kinds

Shop A

6 eggs cost \$1.50

Shop B

12 eggs cost \$2.40

Shop A sells eggs at \$1.50 per half dozen whereas Shop B sells eggs of the same size and quality at \$2.40 per dozen. Which shop should we buy the eggs from?

Solution

Method 1:

Shop A:

6 eggs cost \$1.50 $\div 6$

1 egg costs $\frac{\$1.50}{6}$ $\div 6$

$= \$0.25$

Shop B:

12 eggs cost \$2.40 $\div 12$

1 egg costs $\frac{\$2.40}{12}$ $\div 12$

$= \$0.20$

$\therefore$ we should buy the eggs from Shop B.

Big Idea

Proportionality

In Worked Example 9, two quantities (the number of eggs and their cost) vary proportionally. When one quantity is divided by 6 (or multiplied by $\frac{1}{6}$), we can also divide the other quantity by 6. We say that the cost of the eggs is directly proportional to the number of eggs. We also note that when two quantities are proportional to each other, the rate of change of one quantity with respect to the other quantity is a constant. So, in **Method 2**, we can also use this constant rate for comparison.

Percentages (Sec 1 Textbook p. 19)

Worked Example 13

PPT *

PPT *

Simple reverse percentage problem

In a student council, 30% of the students walk to school. If 48 students walk to school, calculate the number of students in the student council.

Solution

Method 1:

30% of the students = 48

÷ 30

1% of the students = $\frac{48}{30}$

× 100

100% of the students = $\frac{48}{30} \times 100$ ------- (*)

= 160

∴ number of students in student council = 160

Big Idea

Proportionality

In **Method 1**, we observe that when one quantity is divided by 30, the other quantity is also divided by 30; and when one quantity is multiplied by 100, the other quantity is also multiplied by 100. We say that the two quantities vary with each other proportionally.

Equivalent Ratios (Sec 1)

- When there is a need to simplify a ratio, e.g. 15 : 10

$$15 : 10$$

$\div 5$ $\qquad$ $\div 5$

$$3 : 2$$

- Not unitary method but can still use proportional reasoning

Similar Triangles (Sec 2)

- If two triangles are similar, ratios of corresponding sides are equal
- Corresponding sides are **(directly) proportional** to each other
- Video on application of similar triangles in hospitals can be found here: http://math.nie.edu.sg/bwjyeo/videos

Trigonometry (Sec 2)

- Activity: Draw a few right-angled triangles with an angle 30° and measure length of opposite side and of hypotenuse for each triangle
- Observe that $\frac{\text{opp}}{\text{hyp}}$ is equal for all the triangles. Why?
- All the above triangles are similar, so lengths of corresponding sides are **(directly) proportional** to each other
- Lead to definition of $\sin x = \frac{\text{opp}}{\text{hyp}}$

Statistics

- Sec 1: For statistical graphs (e.g. bar chart, pie chart), length of bar, or sector angle/area, is **(directly) proportional** to frequency
- Sec 2: For histogram, area of column is **(directly) proportional** to frequency (but because widths of intervals are equal, length of column is proportional to frequency)
- Upper Sec: For box plot, length of box is **inversely related (not inversely proportional)** to density of data

Recognising Proportional Situations

- Many students fail to distinguish proportional situations from absolute comparison situations, so need to **teach them how to recognise proportional situations** (Behr, Harel, Post & Lesh,1992; Van Dooren, De Bock, Hessels, Janssens, & Verschaffel, 2005)
- By checking that **rate $\frac{y}{x}$ is** _____
- But after teaching them, some may go to the other extreme: is this a proportional situation?

If 30 musicians take 30 minutes to play Beethoven's Fifth Symphony, how long does it take 60 musicians to play the same symphony?

Behr, M., Harel, G., Post, T., & Lesh, R. (1992). Rational number, ratio, and proportion. In D. Grouws (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 296–333). New York: McMillan.

Van Dooren, W., De Bock, D., Hessels, A., Janssens, D., & Verschaffel, L. (2005). Not everything is proportional: effects of age and problem type on propensities for overgeneralisation. *Cognition and Instruction, 23*, 57–86.

Conclusion

- _____ among concepts of ratio, rate and proportion
  - You have to decide what to tell your students and when
- Proportionality is common in maths and in real life: most people may not use equality of ratios or form equations to solve proportional problems, but they actually use **proportional reasoning** (via Unitary Method) to solve
  - Hopefully, you can find ample opportunities in textbooks to highlight big idea about proportionality, that it is **everywhere**